Meridian Postgres HA Architecture Review (2024)

1. Topology

The Meridian transactional tier runs Postgres 16 in a primary/replica pair. db-primary-2 serves all writes and synchronous-committed reads; db-replica-1 streams WAL over a dedicated replication VLAN with a target lag of under 5 seconds. All application traffic reaches Postgres through pgbouncer in transaction pooling mode, with a pool size of 40 per application node.

A third node, db-standby-3, is a delayed replica (recovery_min_apply_delay = 4h) kept as a fat-finger insurance policy: it can be promoted to recover from destructive DDL or bad migrations without going to backups. See the runbook 'Restoring Postgres from Backup with Point-in-Time Recovery' for full PITR when the delayed replica is not enough.

Connection budget: payments-api gets 12 pooled connections, auth-svc 8, search-indexer 6, batch and analytics workloads 8, leaving headroom of 6 connections for operators.

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2. Failover design

Failover is deliberately manual. The on-call promotes db-replica-1 using pg_ctl promote after verifying replication lag is zero and the primary is confirmed dead or isolated - the exact sequence is codified in the runbook 'Failing Over Postgres from db-primary-2 to the Streaming Replica'. Automatic failover was evaluated (Patroni, pg_auto_failover) and rejected for now: our failure history is dominated by network partitions where automation risks split-brain, and the business tolerates the 5-10 minute manual RTO.

pgbouncer is the failover pivot point: repointing its single upstream entry moves every application at once, and its admin console PAUSE/RESUME gives a clean connection drain. DNS is intentionally NOT used for database failover; TTL caching burned us in the 2023 incident and the SOC2 auditors flagged untracked DNS changes.

RPO: zero for synchronous commits. RTO target: 10 minutes, measured 7m40s in the 2024-08 game day.

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3. Capacity and growth

Current write volume peaks at 3,400 TPS during the 14:00-16:00 UTC window, about 41% of the tested ceiling on the current instance class. Storage grows roughly 18 GB/month, dominated by the events and audit_log tables; both are candidates for partitioning in Q1.

The replica currently absorbs read-only analytics traffic between 02:00 and 05:00 UTC. If analytics grows beyond that window, the plan of record is a second streaming replica dedicated to reads rather than moving analytics to the primary.

Action items from this review: partition audit_log by month, add a replication-lag SLO panel to the payments dashboard, and rehearse delayed-replica promotion in the next game day.